

Introduction

- What is machine learning?

Machine learning, a branch of artificial intelligence, concerns the construction and study of systems that can learn from data.

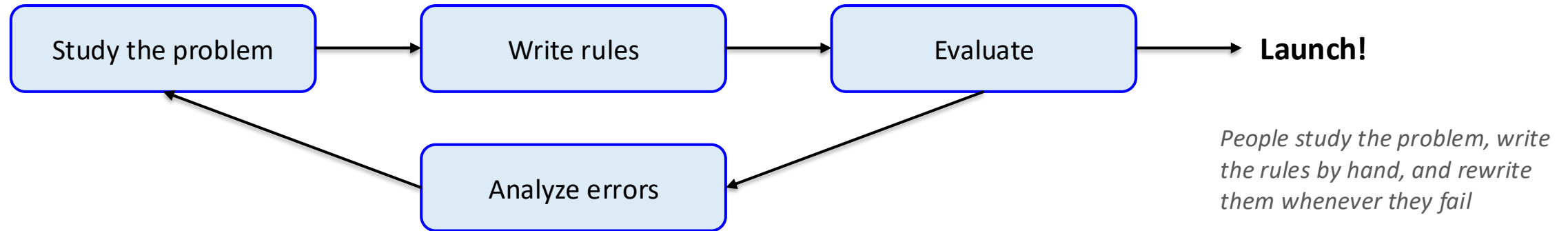


What is machine learning? Two classic definitions

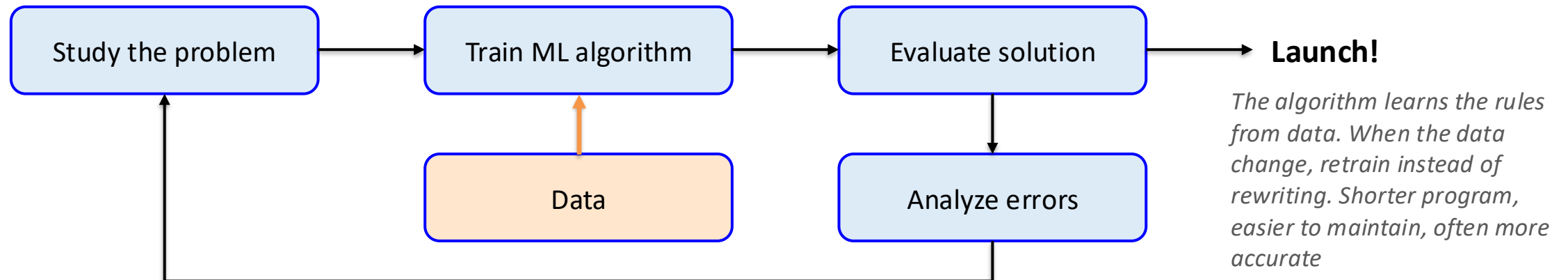
- **Arthur Samuel, 1959:** "the field of study that gives computers the ability to learn without being explicitly programmed"
- **Tom Mitchell, 1997:** "A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E "
- **Example, a spam filter**
 - Task T : flag new emails as spam or not spam
 - Experience E : the training set, thousands of emails already labelled by people. Each one is a training instance
 - Performance P : the fraction of new emails it gets right (accuracy)
- Learning means P goes up as the program sees more of E . Downloading a copy of Wikipedia is not learning: the computer has more data but is no better at any task

Two ways to build that spam filter

Traditional approach

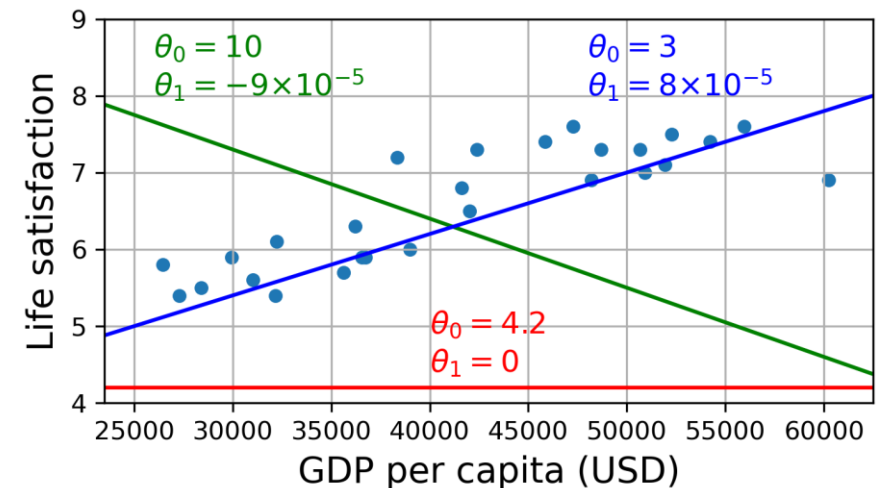
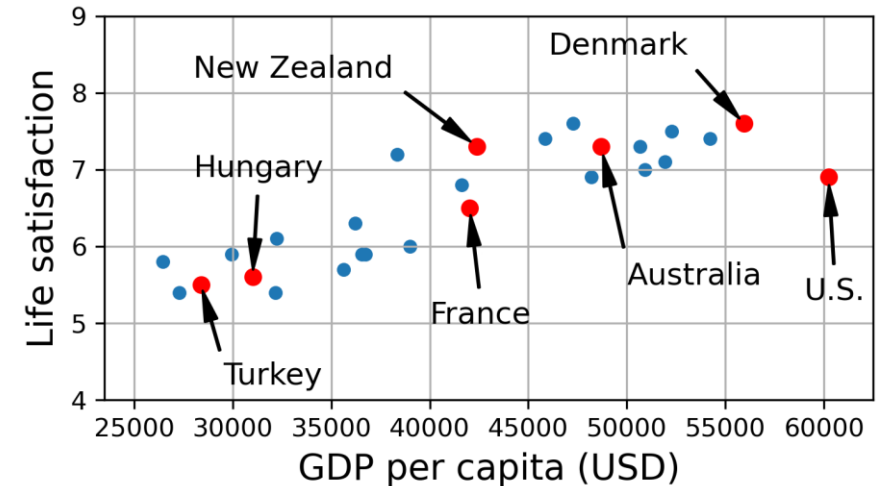


Machine learning approach



Example: does money make people happy?

- **The data:** OECD life satisfaction score (0 to 10) against GDP per capita, one point per country
- **Step 1, study the data.** Messy, but there does seem to be a trend: richer countries report higher satisfaction
- **Step 2, select a model.** The simplest choice is a straight line:
 - $\text{life_satisfaction} = \theta_0 + \theta_1 \times \text{GDP_per_capita}$
- θ_0 and θ_1 are the model parameters. Every pair of values is a different line (bottom figure). Which one should we pick?



Figures regenerated from the author's notebook for Géron, Hands-On Machine Learning, 3rd ed., Ch. 1 (Apache 2.0 code; OECD and World Bank data)

Training the model: pick the parameters that fit best

- **Step 3, train the model.** A **cost function** measures how far the line is from the points, for example the sum of squared vertical distances
 - The learning algorithm searches for the θ_0 and θ_1 that make the cost as small as possible
 - Here it lands on $\theta_0 = 3.75$ and $\theta_1 = 6.78 \times 10^{-5}$
- **Step 4, predict.** Cyprus was not in the training data. Plug in its GDP per capita (37,655 USD) and the model says 6.30 (actual value is 6.2)
- **That is the whole recipe:** study the data, select a model, train it on training data, apply it to new cases (inference)

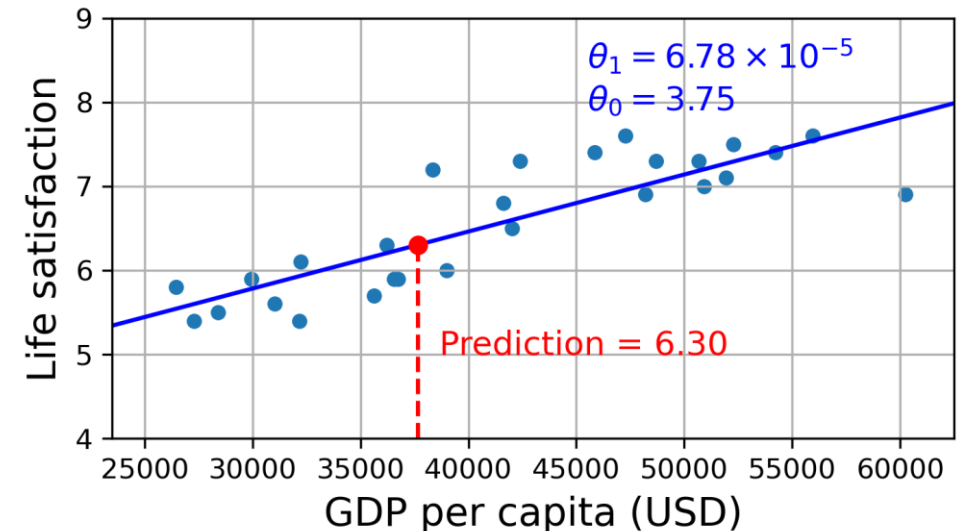
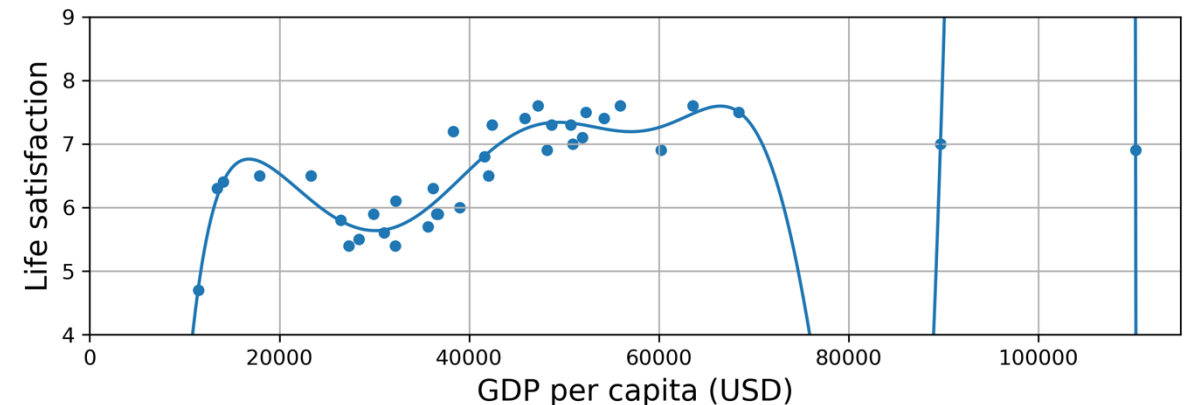
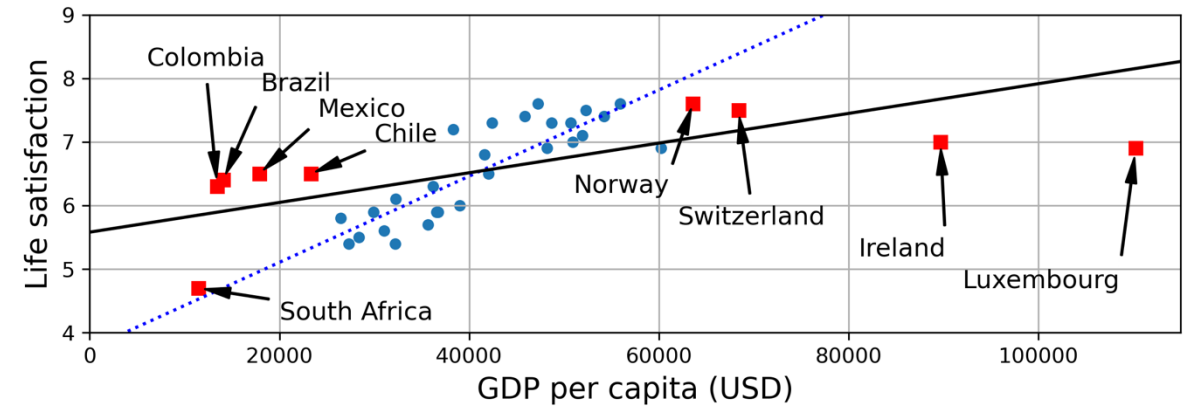


Figure regenerated from the author's notebook for Géron, Hands-On Machine Learning, 3rd ed., Ch. 1 (Apache 2.0 code; OECD and World Bank data)

The catch: it has to work on data it has not seen

- **Top:** add the countries that were left out (red squares) and the line moves a lot. A training set that is not representative gives a misleading model
- **Bottom:** a degree-10 polynomial passes almost through every training point and is useless everywhere else. That is **overfitting**
- The goal is never to fit the training data. It is to **generalize** to new cases
- **Fixes:** more and better data, a simpler model, or regularization (constrain the parameters). We test all of this on held-out data



Figures regenerated from the author's notebook for Géron, Hands-On Machine Learning, 3rd ed., Ch. 1 (Apache 2.0 code; OECD and World Bank data)

AI/ML

ARTIFICIAL INTELLIGENCE

Any technique that enables computers to mimic human behavior



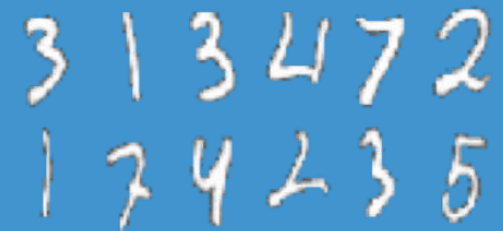
MACHINE LEARNING

Ability to learn without explicitly being programmed



DEEP LEARNING

Extract patterns from data using neural networks



Why use AI for Climate ?

Petabytes of data available from

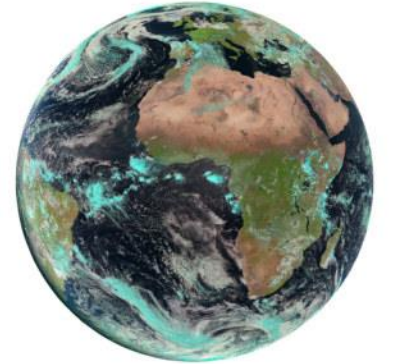
- remote sensing (e.g. satellites)
- climate models
- in-situ observations

Data often contain non-linear dependencies

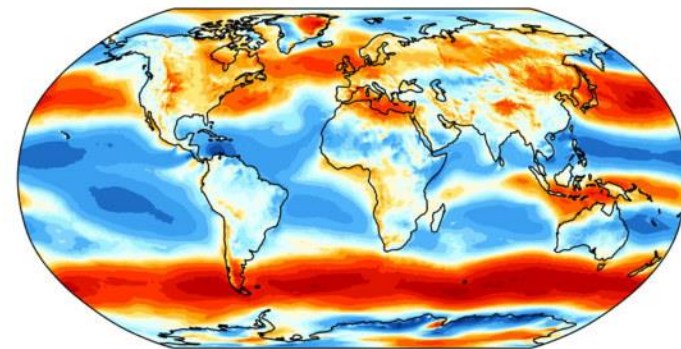
Characteristics of data have analogies to classical ML data (but also key differences!).



Data



Eumetsat: MeteoSat SEVIRI



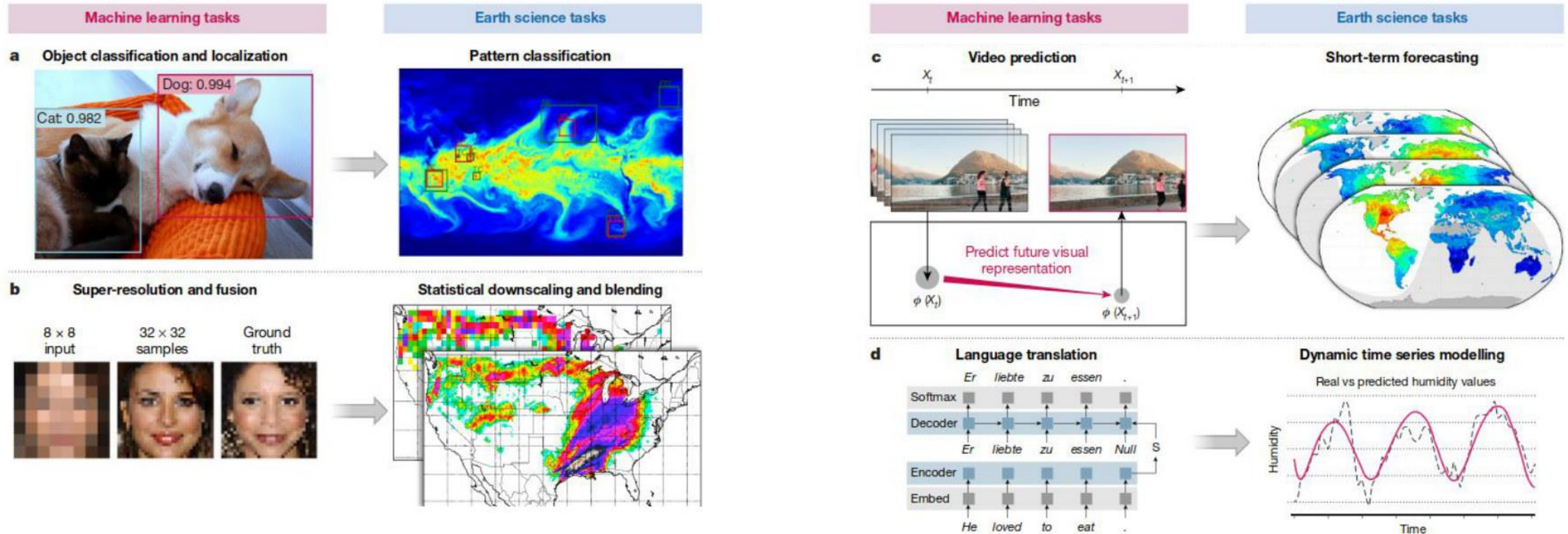
Climate Data Store: ERA5

Why use AI for Climate ?



Research Questions

➡ Analogies to classical ML tasks (predict, classify, detect, etc.)



Why use AI for Climate ?

- **Numerical climate models** and other physics based models in climate science are **very costly to run**.
- **ML-based models** can significantly **reduce cost** and time spent on simulation.
- **Cheaper simulations** enable faster projections and larger ensembles, which **can reduce uncertainty**.

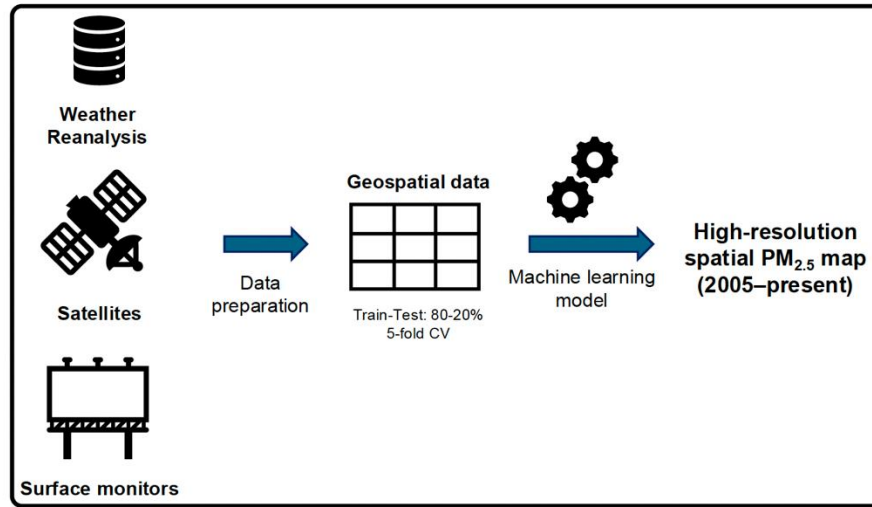


Cost savings

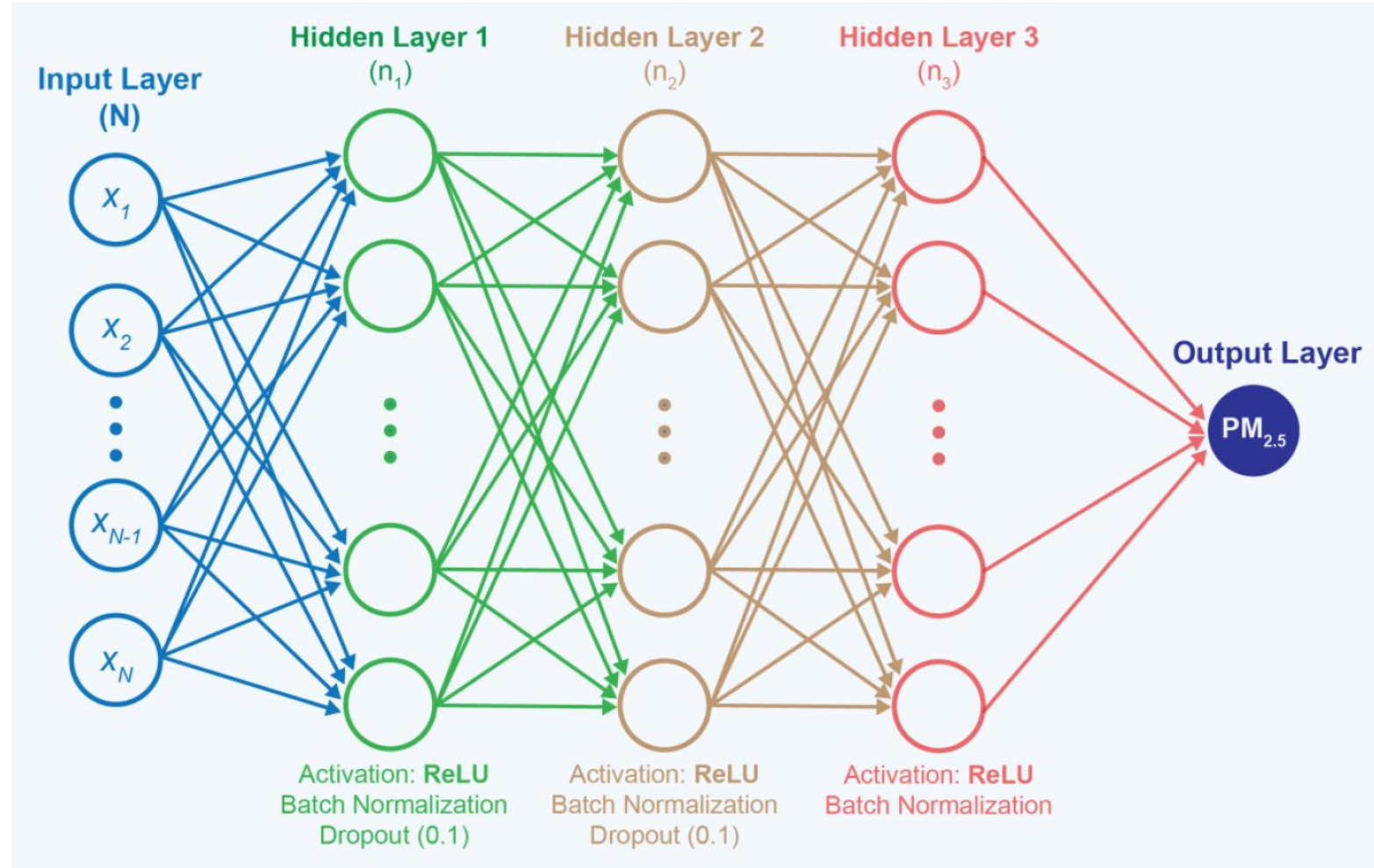
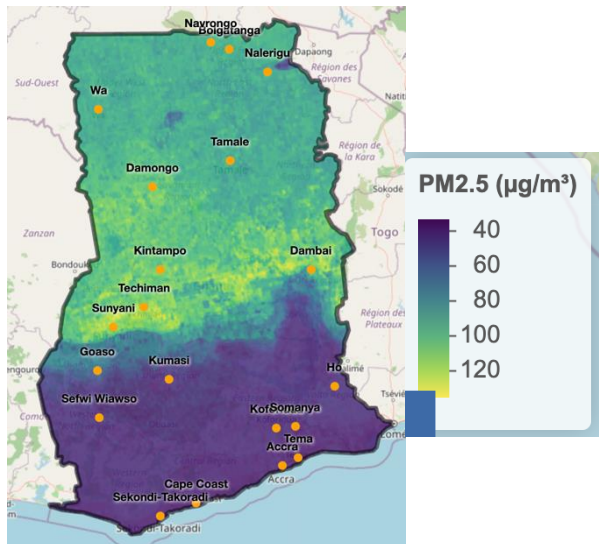


*Swiss National Supercomputing Center:
Piz Daint*

Example of ML in climate from my work



LAMONT-DOHERTY EARTH OBSERVATORY
COLUMBIA CLIMATE SCHOOL



Climate week. Tues September 22, 2:30pm – 5pm

 COLUMBIA
ENGINEERING

 COLUMBIA CLIMATE SCHOOL
Clean Air Toolbox for Cities



Climate Week at Columbia Engineering

Innovation for Climate and Clean Air Progress in the Global South

Tuesday, September 22, 2026 | 2:30pm-6:00pm

 Tang Family Hall, Columbia Engineering Innovation Hub

LAMONT-DOHERTY EARTH OBSERVATORY
 COLUMBIA CLIMATE SCHOOL



https://columbiaengineering.campusgroups.com/CWCE/rsvp_boot?id=375998

What is ML used for, and what do we call things?

- **Regression:** predict a number
 - streamflow tomorrow, CO₂ next year, PM2.5 from a cheap sensor
- **Classification:** predict a category
 - hurricane category, cloud type, healthy vs unhealthy air
- **Clustering:** group similar examples, no labels given
 - climate zones from temperature and rainfall normals
- **Dimensionality reduction:** many variables in, a few out
 - the leading modes of Pacific sea surface temperature
- **Anomaly detection:** flag the unusual examples
 - a drifting sensor in a network

Same thing	Names you will hear for it
The inputs	feature, predictor, covariate, input, independent variable, x
The thing predicted	target, predictand, label, output, response, dependent variable, y
One row of data	example, sample, observation, instance
The fitted function	model, predictor, estimator, learner
Fitting it	training, learning, calibration
Checking it	testing, validation, evaluation

Types of learning

- **Supervised learning**
 - Training data includes desired outputs
- **Unsupervised learning**
 - Training data does not include desired outputs
- **Semi-supervised learning**
 - Training data includes a few desired outputs
- **Reinforcement learning**
 - Rewards from sequence of actions
- **Self-supervised learning**
 - Leveraging data structure – between unsupervised and supervised learning

Specific ML algorithms

Supervised learning

- Linear regression (multiple linear regression, polynomial regression)
 - Regularized via 'ridge regression', 'lasso regression', 'elastic net regression'
- Classifiers
 - Logistic regression, softmax regression
 - Support vector machines
- Decision trees
- Random forests
 - With Gradient Boosting

Unsupervised learning

- Clustering
 - K-means, DBSCAN, Gaussian Mixture Models

Deep learning

- Artificial neural networks
- Convolutional Neural Networks
- Recurrent Neural Networks
- Autoencoders
- Transformers

Supervised learning

Supervised
learning:
given
labeled
examples



label

label₁



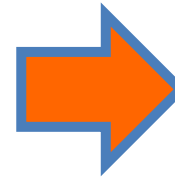
label₃



label₄

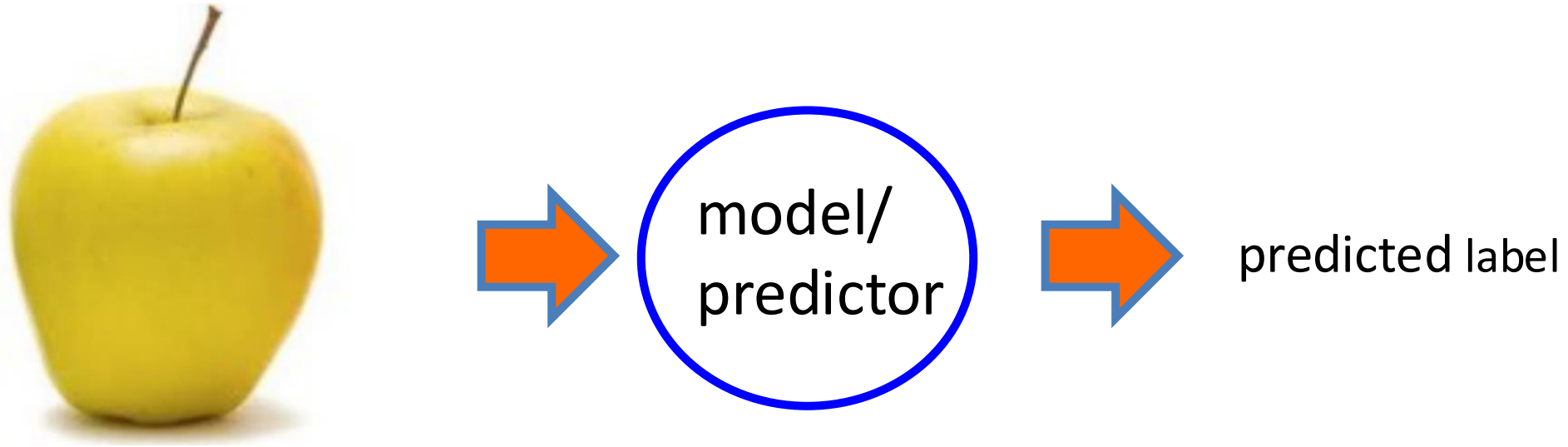


label₅



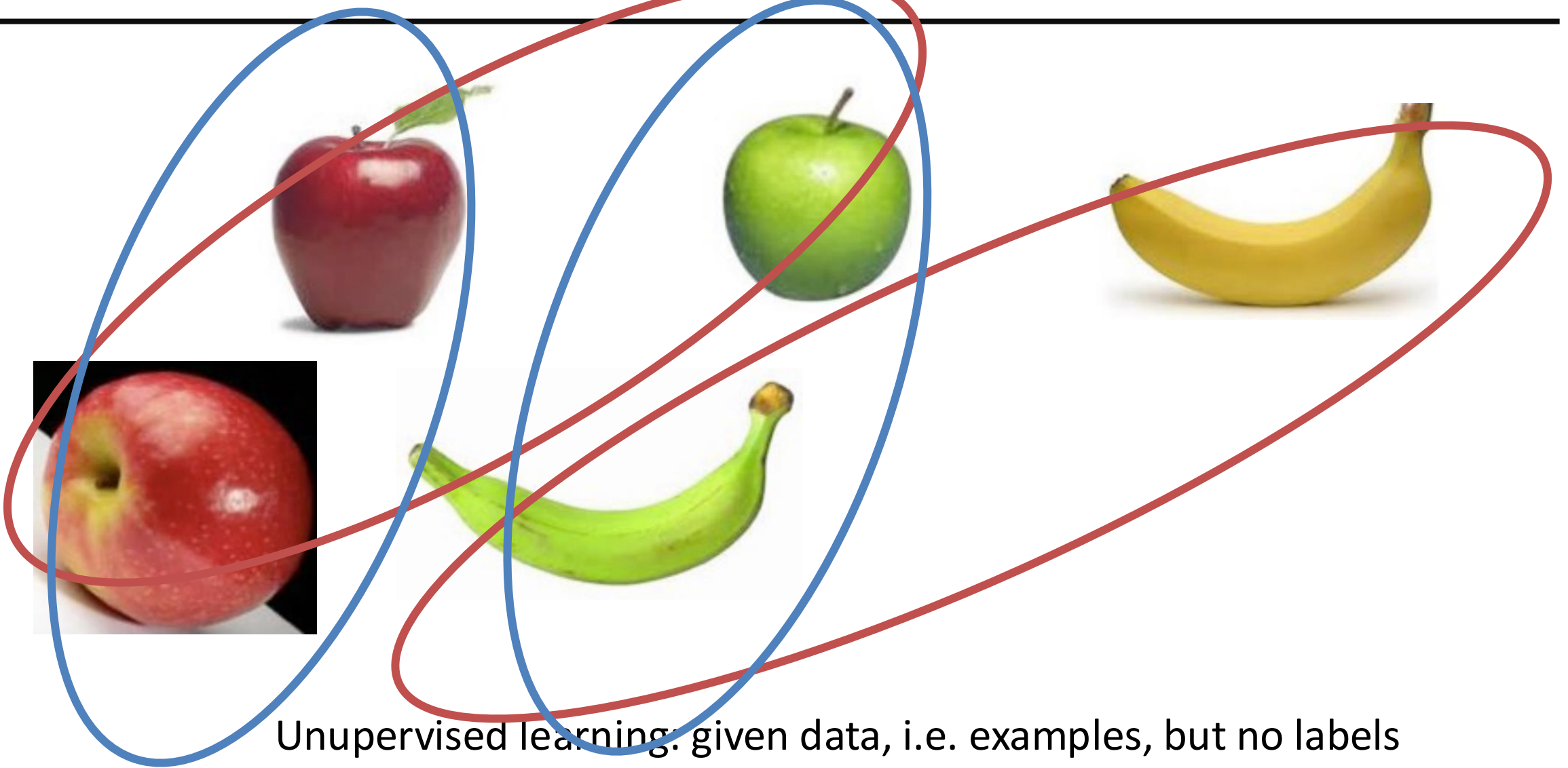
model/
predictor

Supervised learning



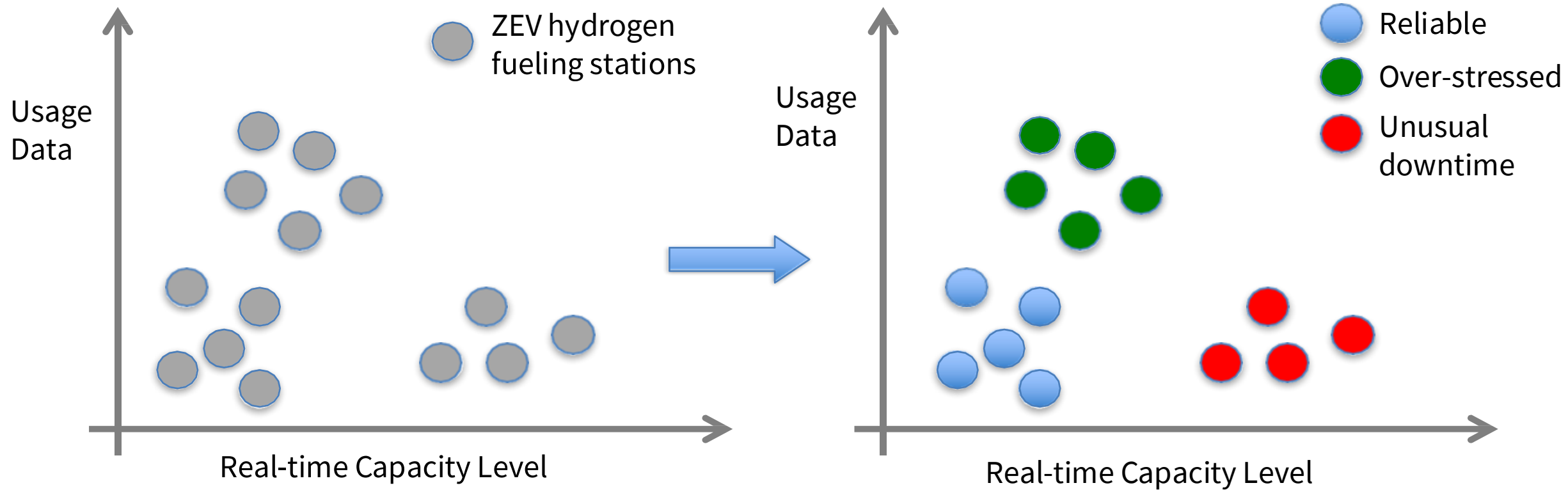
Supervised learning: learn to predict new example

Unsupervised learning



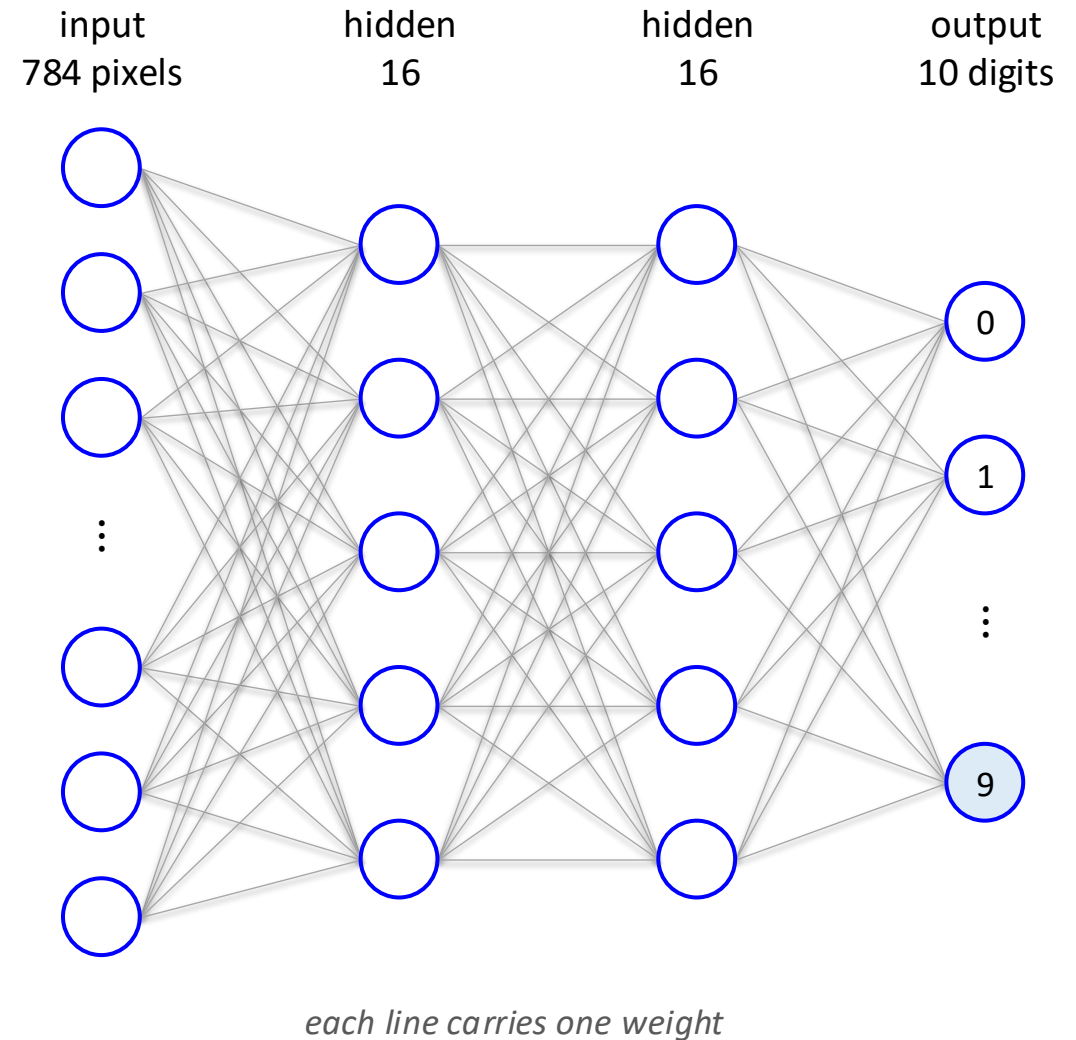
Unsupervised Learning

Finding patterns in unlabeled data



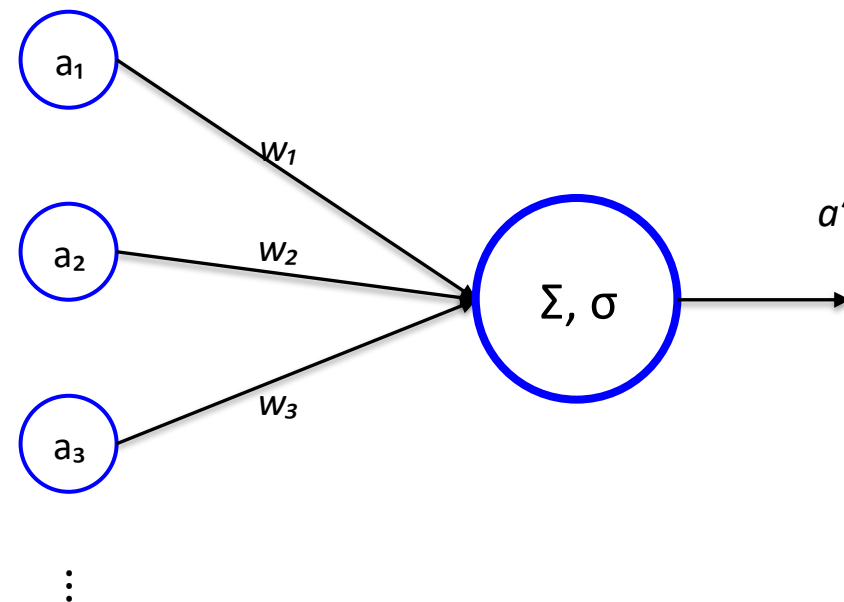
What is a neural network?

- Running example: recognize a handwritten digit
 - A 28×28 image is 784 numbers, the brightness of each pixel (0 to 1)
- A **neuron** is just a thing that holds a number, its **activation**
- Neurons are arranged in layers
 - Input layer: 784 neurons, one per pixel
 - Hidden layers: here two layers of 16
 - Output layer: 10 neurons, one per digit. The brightest one is the answer
- The whole network is just a function: 784 numbers in, 10 numbers out



Inside one neuron

- Each neuron looks at every activation in the previous layer
- Multiply each by a **weight**, add them up, then add a **bias**
 - Weight: how much an input matters, and in which direction
 - Bias: how big the sum must be before the neuron switches on
- Squash the result with an **activation function** (sigmoid then, ReLU now)
- A whole layer at once: $a' = \sigma(Wa + b)$
- The digit network has 13,002 weights and biases, "13,002 knobs and dials that can be tweaked"

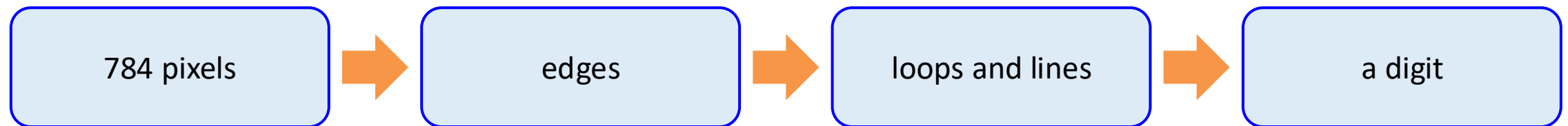


$$a' = \sigma(w_1a_1 + w_2a_2 + w_3a_3 + \dots + b)$$

weighted sum, plus bias, then squashed

Learning means finding the weights

- The hope: each layer builds a little more structure than the last
- Nobody sets 13,002 numbers by hand. **Learning** is getting the computer to find a good setting for all of them
 - Show it labelled examples, compare its output with the right answer, nudge the weights, repeat
 - That is supervised learning again, just with a much bigger model (Weeks 9 to 11)
- Same recipe for our data: satellite pixels or grid points in, a class or a number out



What the layers might learn, from raw input to answer

[Based on 3Blue1Brown, "But what is a neural network?" by Grant Sanderson](#)

First up: quick Python review

Scientific Python “Ecosystem”



Inspiration: Stephan Hoyer, Jake Vanderplas
(SciPy 2015)

HPC in this class

- We will be using a LDEO-based HPC called “chopin”
 - Purchased in 2017, used for research for ~6 years
 - 32 CPUs, 3.2 GHz
 - 256 GB memory per CPU
 - Gigabit ethernet connection
 - Ubuntu 22.04
- JupyterHub
 - A unified coding environment



Important links

- Class jupyterhub
 - <https://chopin.ideo.columbia.edu:8441/jupyterhub/>
 - Log in with your username and password

Gaining access to the jupyterhub

- Look for an email from Dr. Gus Correa with your “chopin” username/password
 - Change your password to something unique and strong immediately!
 - 10 GB home directory
 - 50 GB data directory
- You need to be on the Columbia (or LDEO) network or VPN to connect
- The accounts will be closed after the course ends (by February 2027)